MATH 225

Applied Linear Algebra II

Course Notes

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Lecture 1: General Vector Spaces and Subspaces

Intuition. In previous courses, a "vector" was strictly an arrow in space (with magnitude and direction) or a column of numbers. In Applied Linear Algebra II, we generalize this concept entirely. A vector space is simply a mathematical sandbox. It is any collection of objects (whether they are matrices, polynomials, functions, or coordinates) that strictly obey a specific set of 10 rules. If they obey these rules, they are officially "vectors," and all the powerful theorems of linear algebra apply to them.

The first five axioms govern how addition works internally, and the next five govern how scaling interacts with the vectors.

1.1 Vector Spaces and the Ten Axioms

Definition 1.1 (Vector Space)

A **vector space over** $\mathbb{R}$ is a set V together with two operations, **addition** $\vec{x}+\vec{y}$ and **scalar multiplication** $c\vec{x}$, such that for all $\vec{x},\vec{y},\vec{z}\in V$ and all $c,d\in\mathbb{R}$ the following ten axioms hold.

Vector Addition Axioms

V1 Closure under Addition: $\vec{x}+\vec{y}\in V$

V2 Commutativity: $\vec{x}+\vec{y}=\vec{y}+\vec{x}$

V3 Associativity: $(\vec{x}+\vec{y})+\vec{z}=\vec{x}+(\vec{y}+\vec{z})$

V4 Additive Identity: there is a $\vec{0}\in V$ with $\vec{x}+\vec{0}=\vec{x}$ for all $\vec{x}\in V$

V5 Additive Inverse: for each $\vec{x}\in V$ there is a $-\vec{x}\in V$ with $\vec{x}+(-\vec{x})=\vec{0}$

Scalar Multiplication Axioms

V6 Closure under Scalar Multiplication: $c\vec{x}\in V$

V7 Vector Distributivity: $c(\vec{x}+\vec{y})=c\vec{x}+c\vec{y}$

V8 Scalar Distributivity: $(c+d)\vec{x}=c\vec{x}+d\vec{x}$

V9 Associativity: $(cd)\vec{x}=c(d\vec{x})$

V10 Multiplicative Identity: $1\vec{x}=\vec{x}$

Remark. A set on its own is *not* a vector space. We first have to say what "addition" and "scalar

multiplication" mean on V, and only then check the ten axioms. The same set can be a vector space under one pair of operations and fail under another.

Example 1.2 (Checking All Ten Axioms)

Let $V = \mathbb{R}_{>0}$ be the set of positive real numbers, with "addition" and "scalar multiplication" defined by

$$x \oplus y = xy, \qquad c \odot x = x^c.$$

Show that V is a vector space.

Solution. We go through the axioms one at a time. Throughout, $x, y, z > 0$ and $c, d \in \mathbb{R}$.

V1	$x \oplus y = xy > 0$	so $x \oplus y \in V$
V2	$x \oplus y = xy = yx = y \oplus x$	
V3	$(x \oplus y) \oplus z = (xy)z = x(yz) = x \oplus (y \oplus z)$	
V4	$x \oplus 1 = x \cdot 1 = x$	so the zero vector is 1
V5	$x \oplus \frac{1}{x} = 1$	so $-x = \frac{1}{x} \in V$
V6	$c \odot x = x^c > 0$	so $c \odot x \in V$
V7	$c \odot (x \oplus y) = (xy)^c = x^c y^c = (c \odot x) \oplus (c \odot y)$	
V8	$(c + d) \odot x = x^{c+d} = x^c x^d = (c \odot x) \oplus (d \odot x)$	
V9	$(cd) \odot x = x^{cd} = (x^d)^c = c \odot (d \odot x)$	
V10	$1 \odot x = x^1 = x$	

All ten axioms hold, so V is a vector space. Notice that the "zero vector" here is the number 1, and the "negative" of x is $\frac{1}{x}$. The zero vector is just whatever element acts as the additive identity, it does not have to be the number 0.

Showing a Set Is Not a Vector Space

To show something *is* a vector space, all ten axioms have to hold. To show something is *not* a vector space, we only need **one** axiom to fail, and we prove it fails with a specific counterexample (actual numbers).

Example 1.3 (The Integers)

Let $V = \mathbb{Z}$ with the usual addition and multiplication. Is V a vector space over $\mathbb{R}$?

Solution. No. Closure under scalar multiplication (V6) fails. Take $c = \frac{1}{2}$ and $x = 1 \in \mathbb{Z}$:

$$cx = \tfrac{1}{2} \notin \mathbb{Z}.$$

Scaling a vector in V produced something outside V, so $\mathbb{Z}$ is not a vector space over $\mathbb{R}$.

Example 1.4 (A Strange Scalar Multiplication)

Let $V = \mathbb{R}^2$ with the usual addition, but with scalar multiplication defined by

$$c \odot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} cx \\ 0 \end{bmatrix}.$$

Is V a vector space?

Solution. No. The multiplicative identity axiom (V10) fails:

$$1 \odot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \neq \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

The set $\mathbb{R}^2$ is fine, but with this operation it is not a vector space.

1.2 The Standard Vector Spaces

The elements of a vector space V are called **vectors**. The elements of the field (which for this course is always $\mathbb{R}$) are called **scalars**. These are the four spaces we will use over and over.

Definition 1.5 (Euclidean Space $\mathbb{R}^n$)

$$\mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \;\middle|\; x_1, \ldots, x_n \in \mathbb{R} \right\}$$

with addition and scalar multiplication done component-wise:

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}, \qquad c \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} cx_1 \\ \vdots \\ cx_n \end{bmatrix}.$$

The zero vector is the column of n zeros.

Example 1.6 (Computing in $\mathbb{R}^3$)

Let

$$\vec{x} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \qquad \vec{y} = \begin{bmatrix} 4 \\ 0 \\ -1 \end{bmatrix}.$$

Compute $\vec{x} + \vec{y}$ and $2\vec{x} - \vec{y}$.

Solution. Working one entry at a time:

$$\vec{x} + \vec{y} = \begin{bmatrix} 1+4 \\ -2+0 \\ 3-1 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 2 \end{bmatrix}, \qquad 2\vec{x} - \vec{y} = \begin{bmatrix} 2-4 \\ -4-0 \\ 6+1 \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \\ 7 \end{bmatrix}.$$

Definition 1.7 (Matrix Space $M_{m\times n}(\mathbb{R})$)

$M_{m\times n}(\mathbb{R})$ is the set of all $m \times n$ matrices with real entries, with addition and scalar multiplication done entry by entry:

$$(A+B)_{ij} = (A)_{ij} + (B)_{ij}, \qquad (cA)_{ij} = c\,(A)_{ij}.$$

The zero vector is the zero matrix $O_{m\times n}$, consisting entirely of zeros.

Example 1.8 (Computing in $M_{2\times 2}(\mathbb{R})$)

Let

$$A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}, \qquad B = \begin{bmatrix} 3 & -1 \\ 4 & 2 \end{bmatrix}.$$

Compute $A + B$ and $A - 2B$, and write down the additive inverse of A.

Solution. Entry by entry:

$$A + B = \begin{bmatrix} 4 & 1 \\ 4 & 1 \end{bmatrix}, \qquad A - 2B = \begin{bmatrix} 1-6 & 2+2 \\ 0-8 & -1-4 \end{bmatrix} = \begin{bmatrix} -5 & 4 \\ -8 & -5 \end{bmatrix}.$$

The additive inverse flips every sign, since then $A + (-A) = O_{2\times 2}$:

$$-A = \begin{bmatrix} -1 & -2 \\ 0 & 1 \end{bmatrix}.$$

Definition 1.9 (Polynomial Space $P_n(\mathbb{R})$)

$P_n(\mathbb{R})$ is the set of all polynomials of degree *at most* n with real coefficients:

$$P_n(\mathbb{R}) = \{a_0 + a_1 x + \cdots + a_n x^n \mid a_0, a_1, \ldots, a_n \in \mathbb{R}\},$$

with the operations

$$(p+q)(x) = p(x) + q(x), \qquad (cp)(x) = c\,p(x).$$

In terms of coefficients, we add and scale matching coefficients. The zero vector is the zero polynomial $z(x) = 0 + 0x + \cdots + 0x^n$.

Example 1.10 (Computing in $P_2(\mathbb{R})$)

Let $p(x) = 1 + 2x - x^2$ and $q(x) = 3 - x + 4x^2$. Compute $p+q$ and $2p-q$. Then explain why $P_2(\mathbb{R})$ is defined as degree *at most* 2, not *exactly* 2.

Solution. Combining matching coefficients:

$$p + q = (1+3) + (2-1)x + (-1+4)x^2 = 4 + x + 3x^2,$$

$$2p - q = (2 + 4x - 2x^2) - (3 - x + 4x^2) = -1 + 5x - 6x^2.$$

Now take $r(x) = 1 + x^2$ and $s(x) = 2 - x^2$. Both have degree exactly 2, but

$$r + s = 3,$$

which has degree 0. If we only allowed degree exactly 2, the set would not be closed under addition (V1), and it would not contain the zero polynomial at all. That is why we use "degree at most n."

Definition 1.11 (Function Space $\mathcal{F}(\mathbb{R})$)

$\mathcal{F}(\mathbb{R})$ is the set of all functions $f : \mathbb{R} \to \mathbb{R}$, with

$$(f+g)(x) = f(x) + g(x), \qquad (cf)(x) = c\,f(x).$$

The zero vector is the function that outputs 0 for every x, and $-f$ is the function $(-f)(x) = -f(x)$.

Example 1.12 (Computing in $\mathcal{F}(\mathbb{R})$)

Let $f(x) = \sin x$ and $g(x) = x^2$. Describe $f+g$ and $3f$, and evaluate both at $x = \pi$.

Solution. By the definitions of the operations,

$$(f+g)(x) = \sin x + x^2, \qquad (3f)(x) = 3\sin x.$$

At $x = \pi$:

$$(f+g)(\pi) = 0 + \pi^2 = \pi^2, \qquad (3f)(\pi) = 3 \cdot 0 = 0.$$

The key idea is that each "vector" here is an entire function, and adding two vectors means building a new function point by point.

1.3 Fundamental Properties Derived from the Axioms

Intuition. Because we have redefined "vectors" to be anything from matrices to functions, we cannot blindly assume that basic arithmetic works the same way it does with regular numbers. We must formally prove that things like $0 \cdot \vec{x} = \vec{0}$ are guaranteed strictly by the 10 axioms, without relying on our intuition.

Theorem 1.13 (Uniqueness of the Zero Vector and Inverses)

Let V be a vector space over $\mathbb{R}$. Then:

1. the zero vector $\vec{0}$ in V is unique, and
2. for each $\vec{x} \in V$, the additive inverse $-\vec{x}$ is unique.

Proof. **Proof of (1).** Suppose $\vec{0}$ and $\vec{0}'$ both satisfy V4. Then

$$\begin{aligned} \vec{0} &= \vec{0} + \vec{0}' && (\vec{0}' \text{ is an identity}) \\ &= \vec{0}' + \vec{0} && (\text{by V2}) \\ &= \vec{0}' && (\vec{0} \text{ is an identity}) \end{aligned}$$

So any two zero vectors are equal.

Proof of (2). Suppose $\vec{y}$ and $\vec{z}$ are both inverses of $\vec{x}$, so $\vec{x} + \vec{y} = \vec{0}$ and $\vec{x} + \vec{z} = \vec{0}$. Then

$$\begin{aligned} \vec{y} &= \vec{y} + \vec{0} && (\text{by V4}) \\ &= \vec{y} + (\vec{x} + \vec{z}) && (\vec{z} \text{ is an inverse of } \vec{x}) \\ &= (\vec{x} + \vec{y}) + \vec{z} && (\text{by V3, V2}) \\ &= \vec{0} + \vec{z} = \vec{z} && (\vec{y} \text{ is an inverse, then V2, V4}) \end{aligned}$$

So the inverse of $\vec{x}$ is unique. □

Theorem 1.14 (Elementary Arithmetic Properties of Vector Spaces)

Let V be a vector space over $\mathbb{R}$. For all $\vec{x} \in V$ and all $t \in \mathbb{R}$:

1. $0\vec{x} = \vec{0}$
2. $t\vec{0} = \vec{0}$
3. $(-1)\vec{x} = -\vec{x}$
4. if $t\vec{x} = \vec{0}$, then either $t = 0$ or $\vec{x} = \vec{0}$.

Proof. **Proof of (1).** In the real numbers, $0 + 0 = 0$, so by scalar distributivity (V8)

$$0\vec{x} = (0 + 0)\vec{x} = 0\vec{x} + 0\vec{x}.$$

Since $0\vec{x} \in V$ by closure (V6), its additive inverse $-(0\vec{x})$ exists in V (V5). Adding this inverse to both sides:

$$\begin{aligned} 0\vec{x} + \big(-(0\vec{x})\big) &= \big(0\vec{x} + 0\vec{x}\big) + \big(-(0\vec{x})\big) \\ \vec{0} &= 0\vec{x} + \big(0\vec{x} + (-(0\vec{x}))\big) && \text{(by V5 on left, V3 on right)} \\ \vec{0} &= 0\vec{x} + \vec{0} && \text{(by V5)} \\ \vec{0} &= 0\vec{x} && \text{(by V4)} \end{aligned}$$

Proof of (2). By the additive identity axiom, $\vec{0} + \vec{0} = \vec{0}$. Multiplying by t and using vector distributivity (V7):

$$t\vec{0} = t(\vec{0} + \vec{0}) = t\vec{0} + t\vec{0}.$$

This is the same situation as in (1). Adding $-(t\vec{0})$ to both sides and repeating the same three steps (V5, V3, V4) gives $\vec{0} = t\vec{0}$.

Proof of (3). We show that $(-1)\vec{x}$ behaves exactly as the additive inverse of $\vec{x}$:

$$\begin{aligned} \vec{x} + (-1)\vec{x} &= 1\vec{x} + (-1)\vec{x} && \text{(by V10)} \\ &= \big(1 + (-1)\big)\vec{x} && \text{(by V8)} \\ &= 0\vec{x} = \vec{0} && \text{(by part (1))} \end{aligned}$$

So $(-1)\vec{x}$ is an inverse of $\vec{x}$. By uniqueness of inverses (Theorem 1.13), $(-1)\vec{x} = -\vec{x}$.

Proof of (4). Suppose $t\vec{x} = \vec{0}$. If $t = 0$, the claim holds. If $t \neq 0$, then $t^{-1} \in \mathbb{R}$ exists, and

multiplying both sides by t^{-1}:

$$\begin{aligned} t^{-1}(t\vec{x}) &= t^{-1}\vec{0} \\ (t^{-1}t)\vec{x} &= \vec{0} && \text{(by V9 on left, part (2) on right)} \\ \vec{x} &= \vec{0} && \text{(by } t^{-1}t = 1\text{, then V10)} \end{aligned}$$

So whenever $t \neq 0$, we are forced to have $\vec{x} = \vec{0}$. □

Example 1.15 (Using the Properties)

Let V be a vector space, $\vec{x} \in V$ and $t \in \mathbb{R}$. Show that $(-t)\vec{x} = -(t\vec{x})$.

Solution. Using the theorem above instead of going back to the axioms:

$$\begin{aligned} (-t)\vec{x} &= \big((-1)t\big)\vec{x} && \text{(arithmetic in } \mathbb{R}\text{)} \\ &= (-1)(t\vec{x}) && \text{(by V9)} \\ &= -(t\vec{x}) && \text{(by part (3), applied to } t\vec{x}\text{)} \end{aligned}$$

1.4 Subspaces and the Subspace Test

Intuition. A subspace is simply a smaller, self-contained vector space living inside a larger one (for example, a 2D plane living inside a 3D space). Instead of verifying all 10 axioms to prove a subset is a vector space, we can take a shortcut. Since the subset inherits properties like commutativity and distributivity from the parent space, we only need to check if the subset is a "closed system." If it contains the origin, and doesn't spill out of its boundaries when we add or scale elements, it is a subspace.

Definition 1.16 (Subspace)

Let V be a vector space over $\mathbb{R}$. A subset $U \subseteq V$ is called a **subspace** of V if U is itself a vector space over $\mathbb{R}$ under the addition and scalar multiplication operations inherited from V.

Example 1.17 (Using the Definition Directly)

In $\mathbb{R}^2$, consider the x-axis and a horizontal line shifted up by 1:

$$U = \left\{ \begin{bmatrix} a \\ 0 \end{bmatrix} \,\middle|\, a \in \mathbb{R} \right\}, \qquad L = \left\{ \begin{bmatrix} a \\ 1 \end{bmatrix} \,\middle|\, a \in \mathbb{R} \right\}.$$

Which of these is a subspace of $\mathbb{R}^2$?

Solution. For U: adding and scaling keeps the second entry at 0,

$$\begin{bmatrix} a \\ 0 \end{bmatrix} + \begin{bmatrix} b \\ 0 \end{bmatrix} = \begin{bmatrix} a+b \\ 0 \end{bmatrix} \in U, \qquad c\begin{bmatrix} a \\ 0 \end{bmatrix} = \begin{bmatrix} ca \\ 0 \end{bmatrix} \in U.$$

The zero vector is in U (take $a = 0$), and the inverse of a vector in U is also in U (replace a by $-a$). Every other axiom is an equation that already holds in $\mathbb{R}^2$. So U is a vector space on its own, and hence a subspace.

For L: adding two vectors of L leaves L,

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \notin L,$$

so V1 fails and L is not a subspace.

Notice what we actually had to check for U: the zero vector and the two closure properties. Everything else came for free. That is exactly the shortcut below.

Theorem 1.18 (Subspace Test)

Let V be a vector space over $\mathbb{R}$. A subset $U \subseteq V$ is a subspace of V if and only if:

1. $\vec{0} \in U$, where $\vec{0}$ is the zero vector of V (in particular, $U \neq \emptyset$),
2. for all $\vec{x}, \vec{y} \in U$, $\vec{x} + \vec{y} \in U$ (**closure under addition**),
3. for all $\alpha \in \mathbb{R}$ and $\vec{x} \in U$, $\alpha\vec{x} \in U$ (**closure under scalar multiplication**).

Proof. ($\Longrightarrow$) If U is a subspace, it satisfies all the axioms on its own. In particular it is closed under addition (V1) and scalar multiplication (V6), which gives (2) and (3). For (1), U is non-empty, so pick any $\vec{x} \in U$. By closure $0\vec{x} \in U$, and $0\vec{x} = \vec{0}$ by Theorem 1.14(1). So $\vec{0} \in U$.

($\Longleftarrow$) Assume (1), (2) and (3). We check the ten axioms for U:

- **V1, V6:** these are exactly conditions (2) and (3).
- **V4:** $\vec{0} \in U$ by (1), and $\vec{x} + \vec{0} = \vec{x}$ already holds in V.
- **V5:** for $\vec{x} \in U$, condition (3) with $\alpha = -1$ gives $(-1)\vec{x} \in U$, and $(-1)\vec{x} = -\vec{x}$ by Theorem 1.14(3).
- **V2, V3, V7 to V10:** these are equations that hold for *all* vectors of V, so they hold for the vectors in U.

Thus U is a vector space, and so a subspace of V. □

Remark. Condition (1) is the quickest check, so always do it first. If $\vec{0} \notin U$, you are done: U is not a subspace. But $\vec{0} \in U$ on its own proves nothing, you still need both closure conditions.

Remark. Every vector space V has two **trivial subspaces**: $\{\vec{0}\}$ and V itself. Also, a subspace has to be a *subset* of V, so $\mathbb{R}^2$ is not a subspace of $\mathbb{R}^3$ (its vectors have two entries, not three). More generally, $\mathbb{R}^k$ is not a subspace of $\mathbb{R}^n$ whenever $k \neq n$.

Geometric Picture

In $\mathbb{R}^2$ and $\mathbb{R}^3$, subspaces are exactly the "flat" sets passing through the origin:

- subspaces of $\mathbb{R}^2$: $\{\vec{0}\}$, lines through the origin, and $\mathbb{R}^2$;
- subspaces of $\mathbb{R}^3$: $\{\vec{0}\}$, lines through the origin, planes through the origin, and $\mathbb{R}^3$.

Anything curved (like a parabola), anything shifted off the origin (like the line $y = x + 1$), or anything with an edge (like a quadrant) fails.

1.5 Worked Examples: Subspace Analysis

How to Run the Subspace Test

To prove U is a subspace:

1. Show the zero vector of V satisfies the condition that defines U.
2. Take two *general* elements of U, write down what membership means, and show their sum satisfies the same condition.
3. Take a *general* element of U and a general scalar α, and show the scaled element satisfies the condition.

To prove U is not a subspace: find *one* condition that fails, using specific numbers as a counterexample.

Example 1.19 (Polynomial with Root Constraint)

Let $V = P_2(\mathbb{R})$ and

$$U = \{p(x) \in P_2(\mathbb{R}) \mid p(3) = 0\}.$$

Prove that U is a subspace of $P_2(\mathbb{R})$.

Solution. We test the three conditions of the Subspace Test.

Zero vector. The zero polynomial $z(x) = 0 + 0x + 0x^2$ gives $z(3) = 0$, so $z \in U$.

Closure under addition. Let $p, q \in U$, so $p(3) = 0$ and $q(3) = 0$. Then

$$(p+q)(3) = p(3) + q(3) = 0 + 0 = 0,$$

so $p + q \in U$.

Closure under scalar multiplication. Let $p \in U$ and $\alpha \in \mathbb{R}$. Then

$$(\alpha p)(3) = \alpha\, p(3) = \alpha \cdot 0 = 0,$$

so $\alpha p \in U$.

All three conditions hold, so U is a subspace of $P_2(\mathbb{R})$.

Example 1.20 (Traceless Matrices)

Let $V = M_{2\times 2}(\mathbb{R})$ and

$$W = \{A \in M_{2\times 2}(\mathbb{R}) \mid \operatorname{tr}(A) = 0\},$$

where the trace is the sum of the main diagonal:

$$\operatorname{tr}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = a + d.$$

Show that W is a subspace of $M_{2\times 2}(\mathbb{R})$.

Solution. Zero vector. $\operatorname{tr}(O_{2\times 2}) = 0 + 0 = 0$, so $O_{2\times 2} \in W$.

Closure under addition. Let $A, B \in W$, with

$$A = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}, \qquad B = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix},$$

so $a_1 + d_1 = 0$ and $a_2 + d_2 = 0$. Then

$$\operatorname{tr}(A + B) = (a_1 + a_2) + (d_1 + d_2) = (a_1 + d_1) + (a_2 + d_2) = 0,$$

so $A + B \in W$.

Closure under scalar multiplication. Let $A \in W$ and $\alpha \in \mathbb{R}$. Then

$$\operatorname{tr}(\alpha A) = \alpha a_1 + \alpha d_1 = \alpha(a_1 + d_1) = 0,$$

so $\alpha A \in W$.

Thus W is a subspace of $M_{2\times 2}(\mathbb{R})$.

Example 1.21 (Matrices with Equal Off-Diagonal Entries)

Determine whether

$$S = \left\{ \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \in M_{2\times 2}(\mathbb{R}) \;\middle|\; a_2 = a_3 \right\}$$

is a subspace of $M_{2\times 2}(\mathbb{R})$.

Solution. **Zero vector.** The zero matrix has $a_2 = 0 = a_3$, so $O_{2\times 2} \in S$.

Closure under addition. Let $A, B \in S$, so $a_2 = a_3$ and $b_2 = b_3$. Then

$$A + B = \begin{bmatrix} a_1 + b_1 & a_2 + b_2 \\ a_3 + b_3 & a_4 + b_4 \end{bmatrix},$$

and the off-diagonal entries are equal since $a_2 + b_2 = a_3 + b_3$. So $A + B \in S$.

Closure under scalar multiplication. Let $A \in S$ and $\alpha \in \mathbb{R}$. Then

$$\alpha A = \begin{bmatrix} \alpha a_1 & \alpha a_2 \\ \alpha a_3 & \alpha a_4 \end{bmatrix},$$

and $\alpha a_2 = \alpha a_3$. So $\alpha A \in S$.

Therefore S is a subspace of $M_{2\times 2}(\mathbb{R})$.

Example 1.22 (A Plane Through the Origin in $\mathbb{R}^3$)

Let

$$U = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \;\middle|\; x - 2y + 3z = 0 \right\}.$$

Prove that U is a subspace of $\mathbb{R}^3$.

Solution. **Zero vector.** $0 - 2(0) + 3(0) = 0$, so $\vec{0} \in U$.

Closure under addition. Let

$$\vec{u} = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}, \qquad \vec{v} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

be in U, so $x_1 - 2y_1 + 3z_1 = 0$ and $x_2 - 2y_2 + 3z_2 = 0$. The entries of $\vec{u} + \vec{v}$ satisfy

$$\begin{aligned} (x_1 + x_2) - 2(y_1 + y_2) + 3(z_1 + z_2) &= (x_1 - 2y_1 + 3z_1) + (x_2 - 2y_2 + 3z_2) \\ &= 0 + 0 = 0, \end{aligned}$$

so $\vec{u} + \vec{v} \in U$.

Closure under scalar multiplication. Let $\vec{u} \in U$ and $\alpha \in \mathbb{R}$. The entries of $\alpha\vec{u}$ satisfy

$$(\alpha x_1) - 2(\alpha y_1) + 3(\alpha z_1) = \alpha(x_1 - 2y_1 + 3z_1) = \alpha \cdot 0 = 0,$$

so $\alpha\vec{u} \in U$.

Since it passes the Subspace Test, the plane U is a subspace of $\mathbb{R}^3$.

Example 1.23 (Affine Coefficient Failure)

Let $V = P_2(\mathbb{R})$ and

$$S = \{ax^2 + bx + c \in P_2(\mathbb{R}) \mid a - 2b + c = 3\}.$$

Determine whether S is a subspace of V.

Solution. We always check the zero vector first. The zero polynomial has $a = b = c = 0$, and

$$a - 2b + c = 0 \neq 3.$$

So the zero polynomial is not in S, and condition (1) of the Subspace Test fails immediately. A subspace *must* always contain the zero vector.

Geometrically, in the "space" of coefficients (a, b, c), this is a plane that has been shifted off the origin. Thus S is **not** a subspace of $P_2(\mathbb{R})$.

Example 1.24 (Contains Zero, Still Fails)

Determine whether

$$S = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 \;\middle|\; x_1 = x_2 x_3 \right\}$$

is a subspace of $\mathbb{R}^3$.

Solution. The zero vector passes, since $0 = 0 \cdot 0$, so we look at closure. The condition multiplies two entries together, which is a red flag (it is not linear), so we look for a counterexample. Take

$$\vec{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \qquad \vec{y} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}, \qquad \vec{x} + \vec{y} = \begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix}.$$

Both $\vec{x}$ and $\vec{y}$ are in S since $1 = 1 \cdot 1$ and $2 = 1 \cdot 2$. But for the sum, $3 \neq 2 \cdot 3$, so $\vec{x} + \vec{y} \notin S$. Since S is not closed under addition, it is **not** a subspace of $\mathbb{R}^3$.

Example 1.25 (The First Quadrant)

Let

$$Q = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \;\middle|\; x \geq 0,\ y \geq 0 \right\}.$$

Is Q a subspace of $\mathbb{R}^2$?

Solution. The zero vector is in Q, and Q is even closed under addition (adding two non-negative numbers gives a non-negative number). But scalar multiplication fails:

$$(-1)\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix} \notin Q.$$

So Q is **not** a subspace of $\mathbb{R}^2$. This is why we need all three conditions: passing two of them is not enough.

Lecture 2: Span and Linear Independence

Intuition. If we treat vectors like directional building blocks, the "span" is the set of absolutely everywhere you can possibly reach by combining them. "Linear independence" means there is no redundancy in your building blocks; no vector in the set can be created by mixing the others. If a set spans a space and is linearly independent, it forms a perfect, highly efficient coordinate system for that space.

2.1 Linear Combinations and Span

Definition 2.1 (Linear Combination)

Let V be a vector space over $\mathbb{R}$ and let $S = \{\vec{v}_1, \dots, \vec{v}_k\} \subseteq V$. An expression of the form

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k, \qquad c_1, \dots, c_k \in \mathbb{R},$$

is called a **linear combination** of the vectors in S. The scalars c_i are called its **coefficients**.

Example 2.2 (Writing a Vector as a Linear Combination)

Write

$$\vec{w} = \begin{bmatrix} -6 \\ 7 \\ 1 \end{bmatrix} \quad \text{as a linear combination of} \quad \vec{v}_1 = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 8 \\ -11 \\ 2 \end{bmatrix}.$$

Solution. We want c_1, c_2 with $c_1\vec{v}_1 + c_2\vec{v}_2 = \vec{w}$. Comparing entries gives the system

$$-2c_1 + 8c_2 = -6,$$
$$2c_1 - 11c_2 = 7,$$
$$c_1 + 2c_2 = 1.$$

From the third equation, $c_1 = 1 - 2c_2$. Substituting into the first:

$$-2(1 - 2c_2) + 8c_2 = -6 \implies 12c_2 = -4 \implies c_2 = -\tfrac{1}{3}, \qquad c_1 = 1 + \tfrac{2}{3} = \tfrac{5}{3}.$$

We must check these also satisfy the second equation: $2\left(\frac{5}{3}\right) - 11\left(-\frac{1}{3}\right) = \frac{10}{3} + \frac{11}{3} = 7$. They do, so

$$\vec{w} = \tfrac{5}{3}\,\vec{v}_1 - \tfrac{1}{3}\,\vec{v}_2.$$

Definition 2.3 (Span)

Let $S = \{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq V$. The **span** of S is the set of all possible linear combinations of the vectors in S:

$$\text{Span}(S) = \{c_1\vec{v}_1 + \cdots + c_k\vec{v}_k \mid c_1, \ldots, c_k \in \mathbb{R}\}.$$

If $S = \emptyset$, we define $\text{Span}(\emptyset) = \{\vec{0}\}$.

If $\text{Span}(S) = W$, we say that S **spans** W, or that S is a **spanning set** for W.

Example 2.4 (Elements In and Out of a Span)

Let $S = \{x,\ 1 + x^2\} \subseteq P_2(\mathbb{R})$. Give some elements of $\text{Span}(S)$, and show that $p(x) = 1$ is *not* in $\text{Span}(S)$.

Solution. Any choice of coefficients gives an element of the span:

$$\begin{aligned} 0 &= 0 \cdot x + 0 \cdot (1 + x^2) && c_1 = 0,\ c_2 = 0 \\ x &= 1 \cdot x + 0 \cdot (1 + x^2) && c_1 = 1,\ c_2 = 0 \\ -2 + 3x - 2x^2 &= 3 \cdot x - 2 \cdot (1 + x^2) && c_1 = 3,\ c_2 = -2 \end{aligned}$$

Now suppose $1 \in \text{Span}(S)$. Then for some c_1, c_2,

$$1 = c_1x + c_2(1 + x^2) = c_2 + c_1x + c_2x^2.$$

Matching coefficients, the constant term needs $c_2 = 1$ but the x^2 term needs $c_2 = 0$. That is impossible, so $1 \notin \text{Span}(S)$.

Example 2.5 (Span Membership in $P_2(\mathbb{R})$)

Let $S = \{1,\ 1 - x - x^2,\ 1 - 2x - 2x^2\}$. Is $p(x) = -3 - x + 2x^2$ in $\text{Span}(S)$?

Solution. Expand a general linear combination and group by powers of x:

$$c_1(1) + c_2(1 - x - x^2) + c_3(1 - 2x - 2x^2) = (c_1 + c_2 + c_3) + (-c_2 - 2c_3)x + (-c_2 - 2c_3)x^2.$$

Matching this with $-3 - x + 2x^2$, the x and x^2 coefficients give

$$-c_2 - 2c_3 = -1 \qquad \text{and} \qquad -c_2 - 2c_3 = 2.$$

The same expression cannot equal both -1 and 2, so the system is inconsistent. Therefore $p \notin \text{Span}(S)$.

2.2 Span Is a Subspace

Theorem 2.6 (Span Is a Subspace)

Let V be a vector space over $\mathbb{R}$ and let $S = \{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq V$. Then $\text{Span}(S)$ is a subspace of V.

Proof. We verify the three conditions of the Subspace Test.

Zero vector. Choosing $c_1 = \cdots = c_k = 0$ gives $\vec{0} = 0\vec{v}_1 + \cdots + 0\vec{v}_k \in \text{Span}(S)$.

Closure under addition. Let $\vec{x}, \vec{y} \in \text{Span}(S)$. Then there are scalars $c_i, d_i \in \mathbb{R}$ with

$$\vec{x} = \sum_{i=1}^{k} c_i \vec{v}_i, \qquad \vec{y} = \sum_{i=1}^{k} d_i \vec{v}_i.$$

Adding and using distributivity,

$$\vec{x} + \vec{y} = \sum_{i=1}^{k} c_i \vec{v}_i + \sum_{i=1}^{k} d_i \vec{v}_i = \sum_{i=1}^{k} (c_i + d_i)\vec{v}_i.$$

Each $c_i + d_i \in \mathbb{R}$, so $\vec{x} + \vec{y} \in \text{Span}(S)$.

Closure under scalar multiplication. For any $\alpha \in \mathbb{R}$,

$$\alpha\vec{x} = \alpha \sum_{i=1}^{k} c_i \vec{v}_i = \sum_{i=1}^{k} (\alpha c_i)\vec{v}_i.$$

Each $\alpha c_i \in \mathbb{R}$, so $\alpha\vec{x} \in \text{Span}(S)$. □

Theorem 2.7 (Span Is the Smallest Subspace Containing S)

Let $S = \{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq V$. If W is any subspace of V that contains $\vec{v}_1, \ldots, \vec{v}_k$, then

$$\text{Span}(S) \subseteq W.$$

Proof. Take any $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k \in \text{Span}(S)$. Each $\vec{v}_i \in W$, so each $c_i\vec{v}_i \in W$ by closure under scalar multiplication. Adding them one at a time, the whole sum is in W by closure under addition. So every element of $\text{Span}(S)$ is in W. □

Remark. Together, these two theorems say $\text{Span}(S)$ is the *minimal* subspace containing S: it is a subspace, and every other subspace containing S is at least as big. This also gives a fast way to prove a set is a subspace: if you can write it as a span, you are done.

Example 2.8 (Proving a Subspace by Writing It as a Span)

Let $U = \{p(x) \in P_2(\mathbb{R}) \mid p(1) = 0\}$. Find a spanning set for U, and conclude that U is a subspace.

Solution. Write $p(x) = a + bx + cx^2$. The condition $p(1) = 0$ means $a + b + c = 0$, so $a = -b - c$. Substituting,

$$\begin{aligned} p(x) &= (-b - c) + bx + cx^2 \\ &= b(x - 1) + c(x^2 - 1). \end{aligned}$$

Every element of U is a linear combination of $x - 1$ and $x^2 - 1$, and both of these satisfy $p(1) = 0$. So

$$U = \text{Span}\{x - 1,\ x^2 - 1\},$$

which is a subspace of $P_2(\mathbb{R})$ by Theorem 2.6.

2.3 Worked Examples: Proving a Set Spans a Space

How to Check if S Spans V

To prove that a set of vectors S spans an entire vector space V, you must show that an *arbitrary* (generic) vector in V can always be built using a linear combination of the vectors in S.

1. Write a generic vector of V with letters (e.g. x, y) and set it equal to $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k$.
2. Turn this into a system of linear equations in the c_i.
3. If the system is consistent for *every* choice of the letters, S spans V. If some choice makes it inconsistent, S does not span V.

Example 2.9 (Spanning $\mathbb{R}^2$)

Determine whether

$$S = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$

spans $\mathbb{R}^2$.

Solution. Let (x, y) stand for the entries of an arbitrary vector in $\mathbb{R}^2$. We must determine if there always exist $c_1, c_2 \in \mathbb{R}$ with

$$c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

This vector equation produces the system

$$c_1 + c_2 = x \qquad \text{Equation 1}$$

$$c_1 - c_2 = y \qquad \text{Equation 2}$$

Adding the two equations eliminates c_2, and subtracting Equation 2 from Equation 1 eliminates c_1:

$$2c_1 = x + y \implies c_1 = \frac{x+y}{2}, \qquad 2c_2 = x - y \implies c_2 = \frac{x-y}{2}.$$

Because we can find explicit, real values for c_1 and c_2 regardless of what x and y are, every vector in $\mathbb{R}^2$ can be constructed. Therefore S spans $\mathbb{R}^2$.

Example 2.10 (A Set That Does Not Span $\mathbb{R}^2$)

Determine whether

$$S = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right\}$$

spans $\mathbb{R}^2$.

Solution. Set

$$c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

and row reduce the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 2 & x \\ 2 & 4 & y \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 2 & x \\ 0 & 0 & y - 2x \end{array}\right].$$

The system is consistent only when $y - 2x = 0$. So, for example, the vector with $x = 1$, $y = 0$ cannot be reached, since $0 - 2(1) \neq 0$. Therefore S does **not** span $\mathbb{R}^2$.

Geometrically, the second vector is just 2 times the first, so both point along the same line $y = 2x$, and that line is the whole span.

2.4 Linear Independence

Definition 2.11 (Linear Dependence and Independence)

Let V be a vector space over $\mathbb{R}$ and let $S = \{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq V$.

- S is **linearly dependent** if there exist scalars $c_1, \ldots, c_k \in \mathbb{R}$, **not all zero**, such that
$$c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k = \vec{0}.$$
- S is **linearly independent** if the only solution to
$$c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k = \vec{0}$$
is the **trivial solution** $c_1 = c_2 = \cdots = c_k = 0$.

How to Test for Linear Independence

To test for linear independence, we always set a generic linear combination of the set equal to the zero vector. We then solve for the coefficients c_i.

- If we get a unique solution ($c_1 = c_2 = \cdots = 0$), the set is **independent**.
- If we get free variables (meaning infinitely many non-zero solutions exist), the set is **dependent**, and one vector is just a combination of the others.

Example 2.12 (Independence in $\mathbb{R}^2$)

Determine whether
$$S = \left\{ \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} -3 \\ 5 \end{bmatrix} \right\}$$
is linearly independent.

Solution. Setting
$$c_1 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} -3 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$
gives the system
$$2c_1 - 3c_2 = 0, \qquad 3c_1 + 5c_2 = 0.$$
From the first equation, $c_1 = \frac{3}{2}c_2$. Substituting into the second,
$$\tfrac{9}{2}c_2 + 5c_2 = \tfrac{19}{2}c_2 = 0 \implies c_2 = 0 \implies c_1 = 0.$$
Only the trivial solution exists, so S is **linearly independent**.

Example 2.13 (Linear Dependence in $\mathbb{R}^3$)

Determine whether

$$S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \\ 1 \end{bmatrix} \right\}$$

is linearly independent.

Solution. We set up the dependence equation

$$c_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 3 \\ 7 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

and row reduce its augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & 5 & 7 & 0 \\ 1 & 0 & 1 & 0 \end{array}\right] \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -2 & -2 & 0 \end{array}\right]$$

$$\xrightarrow{R_3 + 2R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Because the third column has no pivot, c_3 is a free variable, so there are infinitely many non-zero solutions. For instance, let $c_3 = -1$. Then:

$$c_2 + c_3 = 0 \implies c_2 = 1 \qquad \text{row 2}$$

$$c_1 + 2c_2 + 3c_3 = 0 \implies c_1 + 2 - 3 = 0 \implies c_1 = 1 \qquad \text{row 1}$$

So

$$1\vec{v}_1 + 1\vec{v}_2 - 1\vec{v}_3 = \vec{0},$$

a non-trivial combination equal to zero. The set is **linearly dependent**. (Notice $\vec{v}_3 = \vec{v}_1 + \vec{v}_2$.)

Example 2.14 (Linear Independence in $P_2(\mathbb{R})$)

Determine whether $S = \{1 + x,\ x + x^2,\ 1 + x^2\}$ is linearly independent in $P_2(\mathbb{R})$.

Solution. Set a linear combination equal to the zero polynomial and group by powers of x:

$$c_1(1 + x) + c_2(x + x^2) + c_3(1 + x^2) = 0 + 0x + 0x^2$$

$$(c_1 + c_3) + (c_1 + c_2)x + (c_2 + c_3)x^2 = 0 + 0x + 0x^2.$$

For polynomials to be equal, their coefficients must match:

$$\begin{aligned} c_1 + c_3 &= 0 \qquad \text{constants} \\ c_1 + c_2 &= 0 \qquad x \text{ coefficients} \\ c_2 + c_3 &= 0 \qquad x^2 \text{ coefficients} \end{aligned}$$

From the first two equations, $c_3 = -c_1$ and $c_2 = -c_1$. Substituting into the third:

$$-c_1 - c_1 = 0 \implies c_1 = 0,$$

and then $c_2 = c_3 = 0$. The trivial solution is the only solution, so S is **linearly independent**.

Example 2.15 (Linear Independence in $M_{2\times 2}(\mathbb{R})$)

Determine whether

$$S = \left\{ \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \right\}$$

is linearly independent.

Solution. Setting a linear combination equal to the zero matrix and adding entry by entry:

$$c_1 \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} c_1 & c_1 + c_2 \\ c_2 + c_3 & c_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

The top-left entry gives $c_1 = 0$ and the bottom-right gives $c_3 = 0$. Then the top-right gives $c_2 = 0$. Only the trivial solution exists, so S is **linearly independent**.

2.5 Properties of Dependent Sets

Theorem 2.16 (Dependence Means Redundancy)

A set $S = \{\vec{v}_1, \ldots, \vec{v}_k\}$ with $k \geq 2$ is linearly dependent if and only if at least one vector $\vec{v}_i$ is a linear combination of the others.

Proof. ($\Longrightarrow$) Suppose $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = \vec{0}$ with some $c_i \neq 0$. Move every other term to the right side and divide by c_i:

$$\vec{v}_i = -\frac{c_1}{c_i}\vec{v}_1 - \cdots - \frac{c_{i-1}}{c_i}\vec{v}_{i-1} - \frac{c_{i+1}}{c_i}\vec{v}_{i+1} - \cdots - \frac{c_k}{c_i}\vec{v}_k.$$

So $\vec{v}_i$ is a linear combination of the others.

($\Longleftarrow$) Suppose $\vec{v}_i = \sum_{j\neq i} a_j\vec{v}_j$. Then

$$a_1\vec{v}_1 + \cdots + a_{i-1}\vec{v}_{i-1} + (-1)\vec{v}_i + a_{i+1}\vec{v}_{i+1} + \cdots + a_k\vec{v}_k = \vec{0}.$$

The coefficient of $\vec{v}_i$ is $-1 \neq 0$, so this is a non-trivial solution and S is dependent. □

Theorem 2.17 (Quick Facts)

Let V be a vector space.

1. Any set containing $\vec{0}$ is linearly dependent.
2. A single vector $\{\vec{v}\}$ is linearly independent if and only if $\vec{v} \neq \vec{0}$.
3. Two vectors $\{\vec{v}_1, \vec{v}_2\}$ are linearly dependent if and only if one is a scalar multiple of the other.

Proof. **(1)** If $\vec{v}_i = \vec{0}$, take $c_i = 1$ and every other coefficient 0. Then $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = 1 \cdot \vec{0} = \vec{0}$, and this is a non-trivial solution.

(2) If $\vec{v} = \vec{0}$, the set is dependent by (1). If $\vec{v} \neq \vec{0}$ and $c\vec{v} = \vec{0}$, then $c = 0$ by Theorem 1.14(4), so only the trivial solution exists.

(3) This is Theorem 2.16 with $k = 2$: one vector is a combination of the other exactly when it is a scalar multiple of it. □

Example 2.18 (Spotting Dependence by Inspection)

Without row reducing, decide whether each set is linearly independent.

$$\text{(a) } \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -3 \\ -6 \end{bmatrix} \right\} \qquad \text{(b) } \{x,\ x^2,\ 0\} \qquad \text{(c) } \{1+x,\ 1-x,\ 2\}$$

Solution. **(a)** Dependent: the second vector is -3 times the first.

(b) Dependent: the set contains the zero polynomial.

(c) Dependent: adding the first two gives the third,

$$(1+x) + (1-x) - 2 = 0,$$

which is a non-trivial combination equal to zero.

2.6 The Importance of Independence

Theorem 2.19 (Unique Representation Theorem)

Let V be a vector space over $\mathbb{R}$ and let $S = \{\vec{v}_1, \ldots, \vec{v}_k\}$ be a spanning set for V. Then every vector in V can be *uniquely* expressed as a linear combination of the vectors in S if and only if S is linearly independent.

Proof. ($\Longrightarrow$) Suppose every vector has a unique representation. The zero vector already has the representation

$$0\vec{v}_1 + \cdots + 0\vec{v}_k = \vec{0}.$$

If $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = \vec{0}$ as well, then by uniqueness of the representation of $\vec{0}$, we must have $c_1 = \cdots = c_k = 0$. Hence S is linearly independent.

($\Longleftarrow$) Suppose S is linearly independent, and let $\vec{v} \in V$. Because S spans V, there is at least one representation $\vec{v} = \sum_{i=1}^{k} c_i\vec{v}_i$. Suppose there is another, $\vec{v} = \sum_{i=1}^{k} d_i\vec{v}_i$. Subtracting the two:

$$\sum_{i=1}^{k} c_i\vec{v}_i - \sum_{i=1}^{k} d_i\vec{v}_i = \vec{0} \implies \sum_{i=1}^{k} (c_i - d_i)\vec{v}_i = \vec{0}.$$

Because S is linearly independent, every coefficient must be zero:

$$c_i - d_i = 0 \implies c_i = d_i \qquad \text{for all } i = 1, \ldots, k.$$

So the two representations are identical, and the linear combination is unique. □

Example 2.20 (Unique vs. Non-Unique Representations)

Compare how $\mathbb{R}^2$ vectors are built from

$$S = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\} \qquad \text{and} \qquad T = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}.$$

Solution. Both sets span $\mathbb{R}^2$. The set S is independent (neither vector is a multiple of the other), so by the earlier spanning example the coefficients are forced:

$$\begin{bmatrix} 5 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad c_1 = \frac{5+1}{2} = 3, \quad c_2 = \frac{5-1}{2} = 2,$$

and there is no other choice.

The set T is dependent (the third vector is the sum of the first two), and the same vector can be

built in more than one way:

$$\begin{bmatrix}2\\3\end{bmatrix} = 2\begin{bmatrix}1\\0\end{bmatrix} + 3\begin{bmatrix}0\\1\end{bmatrix} + 0\begin{bmatrix}1\\1\end{bmatrix} = 1\begin{bmatrix}1\\0\end{bmatrix} + 2\begin{bmatrix}0\\1\end{bmatrix} + 1\begin{bmatrix}1\\1\end{bmatrix}.$$

This is exactly what the theorem predicts: redundancy in the set means representations stop being unique.

Lecture 3: Bases, Dimension, and Expansion

Intuition. A basis is the perfect skeleton of a vector space. It is a set of vectors that spans the entire space (so you can reach absolutely everywhere) but is also linearly independent (so there is absolutely no redundancy in your coordinates). Because of these two properties, a basis provides a perfect, unique coordinate system for any space, ensuring every single vector has exactly one "address."

3.1 Bases

Definition 3.1 (Basis)

Let V be a vector space. A subset $\mathcal{B} = \{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq V$ is called a **basis** for V if it satisfies two conditions:

1. $\mathcal{B}$ is linearly independent (no redundant vectors), and
2. $\text{Span}(\mathcal{B}) = V$ (enough vectors to build the whole space).

Convention: the basis for the trivial subspace $\{\vec{0}\}$ is the empty set $\emptyset$.

Remark. By the Unique Representation Theorem from Lecture 2, a basis is exactly a set where every vector of V can be built in *one and only one* way. Spanning gives "at least one way," independence gives "at most one way."

Example 3.2 (Basis or Not?)

Decide whether each set is a basis for $\mathbb{R}^2$:

$$S = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}, \qquad T = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\}.$$

Solution. **The set S.** In Lecture 2 we showed S spans $\mathbb{R}^2$, and neither vector is a multiple of the other, so S is independent. Both conditions hold, so S **is** a basis.

The set T. T spans $\mathbb{R}^2$ (the first two vectors alone already do), but it is dependent:

$$7\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3\begin{bmatrix} -1 \\ 2 \end{bmatrix} - 2\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 7-3-4 \\ 0+6-6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

So T is **not** a basis. Spanning on its own is not enough; there is a redundant vector.

Standard Bases

Just like the x, y, z axes in geometry, our standard vector spaces have "standard" bases that are the simplest to work with.

Definition 3.3 (Standard Basis)

- $\mathbb{R}^n$**:** let $\vec{e}_i$ be the vector with a 1 in entry i and 0 everywhere else. The standard basis is $\{\vec{e}_1, \ldots, \vec{e}_n\}$. For example, in $\mathbb{R}^3$:

$$\{\vec{e}_1, \vec{e}_2, \vec{e}_3\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

- $P_n(\mathbb{R})$**:** the monomial basis $\{1, x, x^2, \ldots, x^n\}$.
- $M_{m\times n}(\mathbb{R})$**:** let E_{ij} be the matrix with a 1 in position (i, j) and 0 everywhere else. The standard basis is $\{E_{ij} \mid 1 \le i \le m,\ 1 \le j \le n\}$. For example, in $M_{2\times 2}(\mathbb{R})$:

$$\{E_{11}, E_{12}, E_{21}, E_{22}\} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}.$$

Example 3.4 (Writing Vectors in a Standard Basis)

Write each vector as a linear combination of the standard basis of its space:

$$\vec{x} = \begin{bmatrix} 4 \\ -1 \\ 7 \end{bmatrix}, \qquad p(x) = 4 - x + 2x^3, \qquad A = \begin{bmatrix} 3 & -1 \\ 0 & 5 \end{bmatrix}.$$

Solution. With the standard basis, the coefficients are just the entries (or coefficients) you can already see:

$$\vec{x} = 4\vec{e}_1 - 1\vec{e}_2 + 7\vec{e}_3,$$
$$p(x) = 4(1) - 1(x) + 0(x^2) + 2(x^3),$$
$$A = 3E_{11} - 1E_{12} + 0E_{21} + 5E_{22}.$$

This is why standard bases are "standard": reading off the coefficients takes no work.

Example 3.5 (Verifying That $\{1, x, x^2\}$ Is a Basis for $P_2(\mathbb{R})$)

Prove directly from the definition that $\{1, x, x^2\}$ is a basis for $P_2(\mathbb{R})$.

Solution. To rigorously prove this is a basis, we must check both criteria in the definition.

Spanning. Let $p(x) \in P_2(\mathbb{R})$ be an arbitrary polynomial. By definition, it has the form $p(x) = a + bx + cx^2$ where $a, b, c \in \mathbb{R}$. We can write this exactly as a linear combination of our basis elements:

$$p(x) = a(1) + b(x) + c(x^2).$$

Since any $p(x)$ can be constructed this way, $P_2(\mathbb{R}) = \text{Span}\{1, x, x^2\}$.

Linear independence. Suppose a linear combination equals the zero polynomial, which evaluates to 0 for every $x \in \mathbb{R}$:

$$c_1(1) + c_2(x) + c_3(x^2) = 0 \qquad \text{for all } x.$$

Because this equation is an identity (it must hold for *any* real number x), we can strategically plug in values of x:

$$\begin{aligned} x = 0: &\quad c_1 = 0 && \text{kills the } x \text{ and } x^2 \text{ terms} \\ x = 1: &\quad c_2 + c_3 = 0 \\ x = -1: &\quad -c_2 + c_3 = 0 \end{aligned}$$

Adding the last two equations gives $2c_3 = 0$, so $c_3 = 0$, and then $c_2 = 0$. The trivial solution is the only solution, so $\{1, x, x^2\}$ is linearly independent.

Because $\{1, x, x^2\}$ spans $P_2(\mathbb{R})$ and is linearly independent, it is a basis for $P_2(\mathbb{R})$.

3.2 Finding a Basis for a Span

Theorem 3.6 (Removing Redundant Vectors)

Let $\vec{v}_1, \ldots, \vec{v}_k \in V$. If a vector $\vec{v}_i$ is a linear combination of the others, it is essentially "dead weight." Removing it does not alter the span:

$$\text{Span}\{\vec{v}_1, \ldots, \vec{v}_k\} = \text{Span}\{\vec{v}_1, \ldots, \vec{v}_{i-1}, \vec{v}_{i+1}, \ldots, \vec{v}_k\}.$$

Proof. Call the left span W and the right span W'. Every combination of the smaller set is also a combination of the bigger set (just give $\vec{v}_i$ the coefficient 0), so $W' \subseteq W$.

For the other direction, write $\vec{v}_i = \sum_{j \neq i} a_j \vec{v}_j$, and take any $\vec{w} = \sum_{j=1}^{k} c_j \vec{v}_j \in W$. Substituting for $\vec{v}_i$:

$$\vec{w} = \sum_{j \neq i} c_j \vec{v}_j + c_i \sum_{j \neq i} a_j \vec{v}_j = \sum_{j \neq i} (c_j + c_i a_j) \vec{v}_j,$$

which is a combination of the vectors without $\vec{v}_i$. So $W \subseteq W'$, and $W = W'$. □

So to get a basis for $\text{Span}(S)$, we keep throwing away redundant vectors until the set is independent.

Row reduction does this in one go.

How to Find a Basis for $\mathrm{Span}\{\vec{v}_1, \ldots, \vec{v}_k\}$

1. Set up $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = \vec{0}$ and write the coefficient matrix (for vectors in $\mathbb{R}^n$, the columns are just the vectors).
2. Row reduce to REF (or RREF).
3. The **original** vectors that correspond to the pivot columns form a basis. The non-pivot columns are redundant.

Remark. Why does this work? Row operations change the rows, but they never change which linear relations hold between the *columns*. So if column 4 of the RREF equals column 1 plus column 3, the same relation holds between the original vectors. Always take the basis vectors from the *original* set, never from the reduced matrix.

Example 3.7 (A Basis for a Span in $\mathbb{R}^4$)

Let

$$U = \mathrm{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} -2 \\ -2 \\ 2 \\ -6 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 0 \\ 5 \end{bmatrix} \right\}.$$

Find a basis for U and $\dim(U)$.

Solution. To find a basis, we must filter out the redundant vectors. We form the matrix whose columns are the given vectors and row reduce it, which tracks the dependencies between the columns. First we use the leading 1 in column 1 to clear everything below it. Column 2 then has no pivot, so the next pivot is in column 3:

$$\begin{bmatrix} 1 & -2 & 2 & 3 \\ 1 & -2 & 1 & 2 \\ -1 & 2 & 1 & 0 \\ 3 & -6 & 2 & 5 \end{bmatrix} \xrightarrow[R_4-3R_1]{R_2-R_1,\ R_3+R_1} \begin{bmatrix} 1 & -2 & 2 & 3 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & -4 & -4 \end{bmatrix} \xrightarrow{-R_2} \begin{bmatrix} 1 & -2 & 2 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & -4 & -4 \end{bmatrix}$$

$$\xrightarrow[R_4+4R_2]{R_3-3R_2} \begin{bmatrix} 1 & -2 & 2 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1-2R_2} \begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivots are in **column 1** and **column 3**, so columns 2 and 4 are redundant. The RREF even

tells us how:

$$\vec{v}_2 = -2\vec{v}_1, \qquad \vec{v}_4 = \vec{v}_1 + \vec{v}_3.$$

Keeping the **original vectors** from the pivot columns,

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix} \right\}$$

is a basis for U, and $\dim(U) = 2$.

Example 3.8 (A Basis for a Span in $P_2(\mathbb{R})$)

Let $U = \text{Span}\{2 - x + x^2,\ 1 + x - 2x^2,\ 3 - x^2\}$. Find a basis for U.

Solution. Suppose

$$c_1(2 - x + x^2) + c_2(1 + x - 2x^2) + c_3(3 - x^2) = 0.$$

Expanding and grouping by powers of x, each coefficient must be 0:

$$\begin{aligned} 2c_1 + c_2 + 3c_3 &= 0 \qquad \text{constants} \\ -c_1 + c_2 &= 0 \qquad x \text{ terms} \\ c_1 - 2c_2 - c_3 &= 0 \qquad x^2 \text{ terms} \end{aligned}$$

Row reducing the coefficient matrix:

$$\begin{bmatrix} 2 & 1 & 3 \\ -1 & 1 & 0 \\ 1 & -2 & -1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 1 & -2 & -1 \\ -1 & 1 & 0 \\ 2 & 1 & 3 \end{bmatrix} \xrightarrow[R_3 - 2R_1]{R_2 + R_1} \begin{bmatrix} 1 & -2 & -1 \\ 0 & -1 & -1 \\ 0 & 5 & 5 \end{bmatrix}$$

$$\xrightarrow{-R_2} \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 1 \\ 0 & 5 & 5 \end{bmatrix} \xrightarrow[R_1 + 2R_2]{R_3 - 5R_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivots appear in the first and second columns, so the third polynomial is redundant. Indeed, the third column says it is the sum of the first two:

$$(2 - x + x^2) + (1 + x - 2x^2) = 3 - x^2.$$

Keeping the first two original polynomials,

$$\mathcal{B} = \{2 - x + x^2,\ 1 + x - 2x^2\}$$

is a basis for U, and $\dim(U) = 2$.

3.3 Dimension of a Vector Space

Theorem 3.9 (Dimension Is Well Defined)

If $\mathcal{B} = \{\vec{v}_1, \ldots, \vec{v}_k\}$ and $\mathcal{C} = \{\vec{w}_1, \ldots, \vec{w}_\ell\}$ are both bases for a vector space V, then $k = \ell$.

Remark. This is a mathematical guarantee: a vector space always requires the exact same number of basis vectors, regardless of which specific vectors you choose. The idea behind the proof is that if $k < \ell$, then ℓ vectors would be sitting inside the span of only k vectors, which forces them to be dependent. So $\mathcal{C}$ could not be a basis.

Definition 3.10 (Dimension)

If $\mathcal{B} = \{\vec{v}_1, \ldots, \vec{v}_k\}$ is a basis for a vector space V, then the **dimension** of V is the number of vectors in $\mathcal{B}$:

$$\dim(V) = k.$$

In particular, if $V = \{\vec{0}\}$, then $\dim(V) = 0$, since its basis is $\emptyset$. If V has a finite basis, we call V **finite dimensional**.

Example 3.11 (Dimensions of the Standard Spaces)

Find $\dim(\mathbb{R}^n)$, $\dim(P_n(\mathbb{R}))$ and $\dim(M_{m\times n}(\mathbb{R}))$.

Solution. Count the vectors in each standard basis:

$$\begin{aligned} \dim(\mathbb{R}^n) &= n && \text{one } \vec{e}_i \text{ per entry} \\ \dim(P_n(\mathbb{R})) &= n+1 && 1, x, \ldots, x^n\text{; the } +1 \text{ is for the constant term} \\ \dim(M_{m\times n}(\mathbb{R})) &= mn && \text{one } E_{ij} \text{ per matrix entry} \end{aligned}$$

For example, $\dim(P_2(\mathbb{R})) = 3$ and $\dim(M_{2\times 3}(\mathbb{R})) = 6$.

Example 3.12 (Dimension of a Subspace Given by a Condition)

Find a basis and the dimension of

$$S = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{R}^4 \;\middle|\; x_1 + x_2 + x_3 + x_4 = 0 \right\}.$$

Solution. Solve the condition for one variable, $x_4 = -x_1 - x_2 - x_3$, and split the general vector by

its free variables:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ -x_1 - x_2 - x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}.$$

So these three vectors span S. They are also independent: if a combination of them is $\vec{0}$, the first three entries give $x_1 = x_2 = x_3 = 0$ directly. So they form a basis, and

$$\dim(S) = 3.$$

This makes sense: one equation in $\mathbb{R}^4$ removes one degree of freedom, $4 - 1 = 3$.

Example 3.13 (Dimension of the Symmetric 2×2 Matrices)

Find a basis and the dimension of $S = \{A \in M_{2\times 2}(\mathbb{R}) \mid A = A^T\}$.

Solution. Write a general 2×2 matrix with entries a, b, c, d. The condition $A = A^T$ means

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \implies b = c.$$

So every element of S looks like

$$\begin{bmatrix} a & b \\ b & d \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

These three matrices span S, and they are independent (if the combination is the zero matrix, then $a = b = d = 0$ by reading off entries). So they form a basis, and $\dim(S) = 3$.

Counting Vectors

Theorem 3.14 (Too Many, Too Few, Just Right)

Let V be a finite dimensional vector space with $\dim(V) = n$. Then:

1. A set of **more than** n vectors must be linearly dependent. (You have too many vectors, creating guaranteed redundancy.)
2. A set of **fewer than** n vectors cannot span V. (You don't have enough directions to reach every corner of the space.)
3. A set of **exactly** n vectors spans V if and only if it is linearly independent. (If the count is exactly n, passing one test automatically guarantees passing the other.)

Basis Check When $\dim(V) = n$ Is Known

Let $\mathcal{B}$ have k vectors.

1. If $k \neq n$, then $\mathcal{B}$ is **not** a basis. Done.
2. If $k = n$, check only linear independence. Independent means basis, dependent means not a basis.

Example 3.15 (Counting Alone Decides)

Decide whether each set is a basis.

$$\text{(a) } \{1,\ x,\ x^2,\ 2x^2,\ 1-3x\} \subseteq P_2(\mathbb{R}) \qquad \text{(b) } \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\} \subseteq \mathbb{R}^3$$

Solution. **(a)** The set has 5 elements, but $\dim(P_2(\mathbb{R})) = 3$. Because $5 > 3$, there is unavoidable redundancy, so the set is dependent and **not** a basis.

(b) There are only 2 vectors and $\dim(\mathbb{R}^3) = 3$, so this set cannot span $\mathbb{R}^3$. It only describes a 2D plane (the xy-plane), so it is **not** a basis.

Example 3.16 (Right Count, Check Independence)

Is $\mathcal{B} = \{x + x^2,\ 2 - x^2,\ 1 + x + x^2\}$ a basis for $P_2(\mathbb{R})$?

Solution. $\mathcal{B}$ has 3 vectors and $\dim(P_2(\mathbb{R})) = 3$, so we only need independence. Setting

$$c_1(x + x^2) + c_2(2 - x^2) + c_3(1 + x + x^2) = 0$$

and matching coefficients:

$$\begin{aligned} 2c_2 + c_3 &= 0 && \text{constants} \\ c_1 + c_3 &= 0 && x \text{ terms} \\ c_1 - c_2 + c_3 &= 0 && x^2 \text{ terms} \end{aligned}$$

Subtracting the second equation from the third gives $-c_2 = 0$, so $c_2 = 0$. Then the first gives $c_3 = 0$, and the second gives $c_1 = 0$. Only the trivial solution exists, so $\mathcal{B}$ is independent, and therefore a basis for $P_2(\mathbb{R})$. We never had to check spanning.

Example 3.17 (Right Count, Still Fails)

Is

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$$

a basis for $\mathbb{R}^3$?

Solution. The count is right ($3 = \dim(\mathbb{R}^3)$), so we check independence. Adding all three vectors:

$$\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} + \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

This is a non-trivial combination equal to zero, so $\mathcal{B}$ is dependent and **not** a basis.

3.4 Basis Expansion

Lemma 3.18 (Adding a Vector Outside the Span)

Let $C = \{\vec{v}_1, \ldots, \vec{v}_k\}$ be linearly independent. If $\vec{v} \notin \text{Span}(C)$, then $\{\vec{v}_1, \ldots, \vec{v}_k, \vec{v}\}$ is still linearly independent.

Proof. Suppose $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k + a\vec{v} = \vec{0}$. If $a \neq 0$, we could solve

$$\vec{v} = -\frac{c_1}{a}\vec{v}_1 - \cdots - \frac{c_k}{a}\vec{v}_k,$$

which would put $\vec{v}$ in $\text{Span}(C)$, a contradiction. So $a = 0$, and the equation becomes $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = \vec{0}$. Since C is independent, all $c_i = 0$. Only the trivial solution exists. □

Theorem 3.19 (Basis Expansion)

Let V be an n-dimensional vector space. If $C = \{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq V$ is linearly independent with $k < n$, then we can find additional vectors $\vec{v}_{k+1}, \ldots, \vec{v}_n \in V$ such that

$$\{\vec{v}_1, \ldots, \vec{v}_k, \vec{v}_{k+1}, \ldots, \vec{v}_n\}$$

is a basis for V. In other words, you can always "pad" an independent set until it becomes a basis.

Proof. Since $k < n$, C cannot span V (Theorem 3.14(2)), so there is some $\vec{v}_{k+1} \in V$ with $\vec{v}_{k+1} \notin \text{Span}(C)$. By Lemma 3.18, $C \cup \{\vec{v}_{k+1}\}$ is still independent. Repeat this until the set has n vectors. An independent set of exactly n vectors is a basis by Theorem 3.14(3). □

How to Expand to a Basis

1. Start with the independent set C and note how many vectors are missing $(n - k)$.
2. Set a generic vector equal to a combination of C and row reduce to find the condition for it to be *inside* $\text{Span}(C)$.
3. Choose a new vector that intentionally violates that condition, and add it. Repeat if more vectors are needed.

Example 3.20 (Expanding in $\mathbb{R}^3$)

Extend

$$C = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

to a basis for $\mathbb{R}^3$.

Solution. C is linearly independent (the vectors are not scalar multiples). Since $\dim(\mathbb{R}^3) = 3$, we need exactly one more vector, and it must **not** be a linear combination of the vectors in C.

Take a generic vector with entries a, b, c and ask when there exist c_1, c_2 with

$$c_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

Row reducing the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 0 & a \\ 2 & 1 & b \\ 3 & 0 & c \end{array}\right] \xrightarrow[R_3 - 3R_1]{R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 0 & a \\ 0 & 1 & b - 2a \\ 0 & 0 & c - 3a \end{array}\right].$$

Looking at the bottom row, the vector is inside the span exactly when $c - 3a = 0$. Choosing any a, b, c that deliberately break this rule gives a valid extension vector.

Take $a = 2$, $b = 0$, $c = 1$. Since $1 \neq 3(2)$, this vector is outside $\text{Span}(C)$, so by Lemma 3.18 the

expanded set

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

is independent. It has $3 = \dim(\mathbb{R}^3)$ vectors, so it is a full basis for $\mathbb{R}^3$.

Example 3.21 (Expanding in $P_3(\mathbb{R})$)

Extend $C = \{x,\ 1 + x\}$ to a basis for $P_3(\mathbb{R})$.

Solution. The space $P_3(\mathbb{R})$ is 4-dimensional, so we need two more polynomials. Notice that C contains only polynomials of degree at most 1. Any combination of them has degree at most 1, so it can never produce x^2:

$$x^2 \notin \text{Span}\{x,\ 1 + x\}.$$

By Lemma 3.18, $\{x,\ 1 + x,\ x^2\}$ is independent. By the same reasoning, nothing in its span has an x^3 term, so x^3 can be added too. This gives

$$\mathcal{B} = \{x,\ 1 + x,\ x^2,\ x^3\},$$

an independent set of $4 = \dim(P_3(\mathbb{R}))$ polynomials, which is a basis for $P_3(\mathbb{R})$.

3.5 Basis Extension Theorem

Let V be an n-dimensional v.s

$$S = \{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq V$$

1. if $l < n$, then S must be linearly dependent
2. K is strictly $< n$, then S cannot span V
3. if $K = n$ then S spans V if S is linearly independent

The set $B = \left\{ \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \begin{bmatrix} -6 & 0 \\ 4 & 2 \end{bmatrix} \right\} \subseteq M_{2\times 2}(\mathbb{R})$

Lecture 4: Coordinates and Change of Basis

Intuition. A basis is a coordinate system for a vector space. Once we fix a basis, every vector has exactly one "address": the list of coefficients needed to build it. That list is just a column of numbers in $\mathbb{R}^n$, so no matter how abstract the vectors are (polynomials, matrices, functions), we can do all our computations with plain columns. Changing the basis changes the address, and the change of coordinates matrix is the tool that translates addresses from one basis to another.

4.1 Coordinates with Respect to a Basis

Definition 4.1 (Coordinate Vector)

Let $\mathcal{B} = \{\vec{v}_1, \ldots, \vec{v}_n\}$ be a basis for an n-dimensional vector space V, and let $\vec{v} \in V$. Then

$$\vec{v} = c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_n\vec{v}_n = \sum_{i=1}^{n} c_i\vec{v}_i$$

for some $c_1, \ldots, c_n \in \mathbb{R}$. The **coordinate vector of $\vec{v}$ with respect to $\mathcal{B}$** is

$$[\vec{v}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{R}^n.$$

We also call these the **$\mathcal{B}$-coordinates** of $\vec{v}$.

Remark. The coordinate vector $[\vec{v}]_{\mathcal{B}}$ is *unique*. Since $\mathcal{B}$ spans V, the coefficients exist, and since $\mathcal{B}$ is independent, there is only one choice for them (the Unique Representation Theorem from Lecture 2). This is exactly why we need a basis, and not just any spanning set, to define coordinates.

How to Find $[\vec{v}]_{\mathcal{B}}$

1. Set $\vec{v} = c_1\vec{v}_1 + \cdots + c_n\vec{v}_n$.
2. Compare entries (or coefficients) to get a linear system in $c_1, \ldots, c_n$.
3. Solve. The solution, written as a column, is $[\vec{v}]_{\mathcal{B}}$.

Example 4.2 (Coordinates in $\mathbb{R}^2$ Under Three Bases)

Let

$$\vec{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Find the coordinates of $\vec{v}$ with respect to each of the bases

$$\mathcal{E} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}, \qquad \mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}, \qquad \mathcal{C} = \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}.$$

Solution. **Standard basis $\mathcal{E}$.** The coefficients are just the entries:

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \implies [\vec{v}]_{\mathcal{E}} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Basis $\mathcal{B}$. We need

$$c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

The first entry gives $c_1 = 2$, and the second gives $c_1 + c_2 = 3$, so $c_2 = 1$. Hence

$$[\vec{v}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Basis $\mathcal{C}$. We need

$$c_1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

The first entry gives $c_2 = 2$, and the second gives $c_1 + c_2 = 3$, so $c_1 = 1$. Hence

$$[\vec{v}]_{\mathcal{C}} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Same vector, three different addresses: the coordinates depend completely on the basis.

Remark. **Order matters.** In the example above, $\mathcal{B}$ and $\mathcal{C}$ contain exactly the same two vectors, just listed in a different order. As sets they are equal, but

$$[\vec{v}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \neq \begin{bmatrix} 1 \\ 2 \end{bmatrix} = [\vec{v}]_{\mathcal{C}}.$$

So for coordinates, the order of the basis vectors matters, and we do *not* treat $\mathcal{B}$ and $\mathcal{C}$ as the same basis. From now on, every basis is an **ordered basis**: the first coordinate always goes with the first basis vector, the second with the second, and so on.

Example 4.3 (From Coordinates Back to the Vector)

Let $V = P_2(\mathbb{R})$ with basis $\mathcal{B} = \{1 + x,\ 1 - x^2,\ 1 + x + x^2\}$. If

$$[p(x)]_{\mathcal{B}} = \begin{bmatrix} -2 \\ 5 \\ 1 \end{bmatrix},$$

find $p(x)$.

Solution. The coordinates are the coefficients, matched in order with the basis vectors:

$$\begin{aligned} p(x) &= -2(1 + x) + 5(1 - x^2) + 1(1 + x + x^2) \\ &= (-2 + 5 + 1) + (-2 + 1)x + (-5 + 1)x^2 \\ &= 4 - x - 4x^2. \end{aligned}$$

Going from coordinates to the vector is just plugging in. Going the other way takes a little more work, as the next example shows.

Example 4.4 (From the Vector to Its Coordinates)

With the same basis $\mathcal{B} = \{1 + x,\ 1 - x^2,\ 1 + x + x^2\}$, find $[p]_{\mathcal{B}}$ for $p(x) = 4 - x - 4x^2$.

Solution. Set $p(x) = c_1(1 + x) + c_2(1 - x^2) + c_3(1 + x + x^2)$ and group by powers of x:

$$4 - x - 4x^2 = (c_1 + c_2 + c_3) + (c_1 + c_3)x + (-c_2 + c_3)x^2.$$

Matching coefficients gives the system

$$\begin{aligned} c_1 + c_2 + c_3 &= 4 && \text{constants} \\ c_1 + c_3 &= -1 && x \text{ terms} \\ -c_2 + c_3 &= -4 && x^2 \text{ terms} \end{aligned}$$

Subtracting the second equation from the first gives $c_2 = 5$. Then the third gives $c_3 = -4 + 5 = 1$, and the second gives $c_1 = -1 - 1 = -2$. So

$$[p]_{\mathcal{B}} = \begin{bmatrix} -2 \\ 5 \\ 1 \end{bmatrix},$$

which matches the previous example, as it should.

Example 4.5 (Membership and Coordinates in a Subspace of $M_{2\times 2}(\mathbb{R})$)

Let

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 3 \\ 6 & 7 \end{bmatrix} \right\}$$

be a basis for a subspace U of $M_{2\times 2}(\mathbb{R})$. Determine whether

$$A = \begin{bmatrix} 5 & -5 \\ -10 & -19 \end{bmatrix}$$

is in U, and if so, find $[A]_{\mathcal{B}}$.

Solution. $A \in U$ exactly when there exist $c_1, c_2 \in \mathbb{R}$ with

$$\begin{bmatrix} 5 & -5 \\ -10 & -19 \end{bmatrix} = c_1 \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 & 3 \\ 6 & 7 \end{bmatrix}.$$

Comparing the four entries gives the linear system

$$\begin{aligned} c_1 - c_2 &= 5 && \text{top left} \\ 2c_1 + 3c_2 &= -5 && \text{top right} \\ 4c_1 + 6c_2 &= -10 && \text{bottom left} \\ c_1 + 7c_2 &= -19 && \text{bottom right} \end{aligned}$$

From the first equation, $c_1 = 5 + c_2$. Substituting into the second:

$$2(5 + c_2) + 3c_2 = -5 \implies 5c_2 = -15 \implies c_2 = -3, \qquad c_1 = 2.$$

We have four equations but only two unknowns, so we must check the other two:

$$4(2) + 6(-3) = -10 \checkmark, \qquad 2 + 7(-3) = -19 \checkmark.$$

The system is consistent, so $A \in U$, and

$$[A]_{\mathcal{B}} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}.$$

Notice that A lives in $M_{2\times 2}(\mathbb{R})$, but its coordinate vector lives in $\mathbb{R}^2$, because U is only 2-dimensional.

4.2 Coordinates Are Linear

Theorem 4.6 (Linearity of Coordinates)

Let $\mathcal{B} = \{\vec{v}_1, \ldots, \vec{v}_n\}$ be a basis for an n-dimensional vector space V. Then for any $\vec{x}, \vec{y} \in V$ and $s, t \in \mathbb{R}$,

$$[s\vec{x} + t\vec{y}]_{\mathcal{B}} = s\,[\vec{x}]_{\mathcal{B}} + t\,[\vec{y}]_{\mathcal{B}}.$$

Proof. Write $\vec{x} = \sum_{i=1}^{n} a_i \vec{v}_i$ and $\vec{y} = \sum_{i=1}^{n} b_i \vec{v}_i$, so the a_i and b_i are the entries of $[\vec{x}]_{\mathcal{B}}$ and $[\vec{y}]_{\mathcal{B}}$. Then

$$s\vec{x} + t\vec{y} = \sum_{i=1}^{n} sa_i \vec{v}_i + \sum_{i=1}^{n} tb_i \vec{v}_i = \sum_{i=1}^{n} (sa_i + tb_i)\vec{v}_i.$$

By uniqueness of coordinates, the $\mathcal{B}$-coordinates of $s\vec{x} + t\vec{y}$ are exactly $sa_i + tb_i$:

$$[s\vec{x} + t\vec{y}]_{\mathcal{B}} = \begin{bmatrix} sa_1 + tb_1 \\ \vdots \\ sa_n + tb_n \end{bmatrix} = s \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} + t \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = s\,[\vec{x}]_{\mathcal{B}} + t\,[\vec{y}]_{\mathcal{B}}.$$

□

Remark. This says that taking coordinates respects addition and scaling. So any computation in an n-dimensional space V can be done entirely in $\mathbb{R}^n$: convert to coordinates, compute with columns, and convert back.

Example 4.7 (Using Linearity)

In $\mathbb{R}^2$, let

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}, \qquad \vec{x} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \qquad \vec{y} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Find $[2\vec{x} - 3\vec{y}]_{\mathcal{B}}$ in two ways.

Solution. **Using the theorem.** From the first example, $[\vec{x}]_{\mathcal{B}}$ has entries $2, 1$. For $\vec{y}$, the first entry gives $c_1 = 1$ and the second gives $c_1 + c_2 = -1$, so $c_2 = -2$. Then

$$[2\vec{x} - 3\vec{y}]_{\mathcal{B}} = 2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} - 3 \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \end{bmatrix}.$$

Directly. First compute the vector:

$$2\vec{x} - 3\vec{y} = \begin{bmatrix} 4 - 3 \\ 6 + 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 9 \end{bmatrix}.$$

Its $\mathcal{B}$-coordinates satisfy $c_1 = 1$ and $c_1 + c_2 = 9$, so $c_2 = 8$. Both methods give the same answer.

4.3 Change of Coordinates

Intuition. If we know the address of a vector in basis $\mathcal{C}$, we would like a quick way to get its address in basis $\mathcal{B}$ without rebuilding the vector first. Because coordinates are linear, this translation is done by a single matrix.

Definition 4.8 (Change of Coordinates Matrix)

Let $\mathcal{B}$ and $\mathcal{C} = \{\vec{w}_1, \ldots, \vec{w}_n\}$ be two bases for an n-dimensional vector space V. The **change of coordinates matrix from $\mathcal{C}$-coordinates to $\mathcal{B}$-coordinates** is the $n \times n$ matrix whose columns are the $\mathcal{B}$-coordinates of the vectors in $\mathcal{C}$:

$$ {}_{\mathcal{B}}P_{\mathcal{C}} = \Big[\, [\vec{w}_1]_{\mathcal{B}} \;\; [\vec{w}_2]_{\mathcal{B}} \;\; \cdots \;\; [\vec{w}_n]_{\mathcal{B}} \,\Big]. $$

Remark. Read the subscripts from right to left: ${}_{\mathcal{B}}P_{\mathcal{C}}$ takes in $\mathcal{C}$-coordinates (on the right) and outputs $\mathcal{B}$-coordinates (on the left). The subscripts "chain" nicely in the formula below, since the inner $\mathcal{C}$'s line up.

Theorem 4.9 (Changing Coordinates)

Let $\mathcal{B}$ and $\mathcal{C}$ be bases for an n-dimensional vector space V. For every $\vec{x} \in V$,

$$ [\vec{x}]_{\mathcal{B}} = {}_{\mathcal{B}}P_{\mathcal{C}}\, [\vec{x}]_{\mathcal{C}}. $$

Proof. Let $\mathcal{C} = \{\vec{w}_1, \ldots, \vec{w}_n\}$ and write $\vec{x} = c_1\vec{w}_1 + \cdots + c_n\vec{w}_n$, so $[\vec{x}]_{\mathcal{C}}$ has entries $c_1, \ldots, c_n$. Using linearity of coordinates (Theorem 4.6):

$$
\begin{aligned}
[\vec{x}]_{\mathcal{B}} &= [c_1\vec{w}_1 + \cdots + c_n\vec{w}_n]_{\mathcal{B}} \\
&= c_1[\vec{w}_1]_{\mathcal{B}} + \cdots + c_n[\vec{w}_n]_{\mathcal{B}} && \text{linearity} \\
&= \Big[\, [\vec{w}_1]_{\mathcal{B}} \;\; \cdots \;\; [\vec{w}_n]_{\mathcal{B}} \,\Big] \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} && \text{matrix times vector = combination of columns} \\
&= {}_{\mathcal{B}}P_{\mathcal{C}}\, [\vec{x}]_{\mathcal{C}}.
\end{aligned}
$$

□

Example 4.10 (Building a Change of Coordinates Matrix from the Definition)

In $\mathbb{R}^2$, let $\mathcal{E}$ be the standard basis and

$$ \mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}, \qquad \vec{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}. $$

Find ${}_{\mathcal{B}}P_{\mathcal{E}}$, and use it to find $[\vec{v}]_{\mathcal{B}}$.

Solution. We need the $\mathcal{B}$-coordinates of each standard basis vector:

$$\begin{bmatrix}1\\0\end{bmatrix} = 1\begin{bmatrix}1\\1\end{bmatrix} - 1\begin{bmatrix}0\\1\end{bmatrix} \qquad \Longrightarrow\ [\vec{e}_1]_{\mathcal{B}} = \begin{bmatrix}1\\-1\end{bmatrix},$$

$$\begin{bmatrix}0\\1\end{bmatrix} = 0\begin{bmatrix}1\\1\end{bmatrix} + 1\begin{bmatrix}0\\1\end{bmatrix} \qquad \Longrightarrow\ [\vec{e}_2]_{\mathcal{B}} = \begin{bmatrix}0\\1\end{bmatrix}.$$

Putting these in as columns,

$${}_{\mathcal{B}}P_{\mathcal{E}} = \begin{bmatrix}1 & 0\\-1 & 1\end{bmatrix}.$$

Since $[\vec{v}]_{\mathcal{E}}$ is just $\vec{v}$ itself,

$$[\vec{v}]_{\mathcal{B}} = {}_{\mathcal{B}}P_{\mathcal{E}}\,[\vec{v}]_{\mathcal{E}} = \begin{bmatrix}1 & 0\\-1 & 1\end{bmatrix}\begin{bmatrix}2\\3\end{bmatrix} = \begin{bmatrix}2\\1\end{bmatrix},$$

which matches what we found by hand in the first example.

Theorem 4.11 (Inverse of a Change of Coordinates Matrix)

Let $\mathcal{B}$ and $\mathcal{C}$ be bases for an n-dimensional vector space V. Then ${}_{\mathcal{B}}P_{\mathcal{C}}$ is invertible, and

$$\left({}_{\mathcal{B}}P_{\mathcal{C}}\right)^{-1} = {}_{\mathcal{C}}P_{\mathcal{B}}.$$

Proof. Apply Theorem 4.9 twice. For every $\vec{x} \in V$,

$${}_{\mathcal{C}}P_{\mathcal{B}}\ {}_{\mathcal{B}}P_{\mathcal{C}}\,[\vec{x}]_{\mathcal{C}} = {}_{\mathcal{C}}P_{\mathcal{B}}\,[\vec{x}]_{\mathcal{B}} = [\vec{x}]_{\mathcal{C}}.$$

Every column $\vec{a} \in \mathbb{R}^n$ is $[\vec{x}]_{\mathcal{C}}$ for some $\vec{x}$ (take $\vec{x} = a_1\vec{w}_1 + \cdots + a_n\vec{w}_n$), so the matrix ${}_{\mathcal{C}}P_{\mathcal{B}}\ {}_{\mathcal{B}}P_{\mathcal{C}}$ sends every $\vec{a}$ to itself. The only matrix that does this is I. The same argument with $\mathcal{B}$ and $\mathcal{C}$ swapped gives the product in the other order, so the two matrices are inverses. □

How to Compute ${}_{\mathcal{B}}P_{\mathcal{C}}$

Write every vector in standard coordinates. Put the $\mathcal{B}$ vectors on the left and the $\mathcal{C}$ vectors on the right, and row reduce:

$$\left[\begin{array}{ccc|ccc}\vec{b}_1 & \cdots & \vec{b}_n & \vec{w}_1 & \cdots & \vec{w}_n\end{array}\right] \longrightarrow \left[\begin{array}{c|c} I & {}_{\mathcal{B}}P_{\mathcal{C}}\end{array}\right].$$

This works because finding $[\vec{w}_j]_{\mathcal{B}}$ means solving a system whose coefficient matrix has the $\mathcal{B}$ vectors as columns. All n systems share that coefficient matrix, so we solve them all at once.

Shortcut: once you have one direction, the other is the inverse, ${}_{\mathcal{C}}P_{\mathcal{B}} = \left({}_{\mathcal{B}}P_{\mathcal{C}}\right)^{-1}$.

Example 4.12 (Change of Coordinates in $\mathbb{R}^2$)

Consider the two bases

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}, \qquad \mathcal{C} = \left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ -4 \end{bmatrix} \right\}$$

for $\mathbb{R}^2$. Determine ${}_{\mathcal{C}}P_{\mathcal{B}}$ and ${}_{\mathcal{B}}P_{\mathcal{C}}$.

Solution. For ${}_{\mathcal{C}}P_{\mathcal{B}}$ we need the $\mathcal{C}$-coordinates of the $\mathcal{B}$ vectors, so $\mathcal{C}$ goes on the left:

$$\left[\begin{array}{cc|cc} -1 & 5 & 1 & 2 \\ 1 & -4 & 3 & 1 \end{array}\right] \xrightarrow{R_2+R_1} \left[\begin{array}{cc|cc} -1 & 5 & 1 & 2 \\ 0 & 1 & 4 & 3 \end{array}\right] \xrightarrow{-R_1} \left[\begin{array}{cc|cc} 1 & -5 & -1 & -2 \\ 0 & 1 & 4 & 3 \end{array}\right]$$

$$\xrightarrow{R_1+5R_2} \left[\begin{array}{cc|cc} 1 & 0 & 19 & 13 \\ 0 & 1 & 4 & 3 \end{array}\right].$$

Hence

$${}_{\mathcal{C}}P_{\mathcal{B}} = \begin{bmatrix} 19 & 13 \\ 4 & 3 \end{bmatrix}.$$

We can check the first column, which should rebuild the first vector of $\mathcal{B}$:

$$19 \begin{bmatrix} -1 \\ 1 \end{bmatrix} + 4 \begin{bmatrix} 5 \\ -4 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}. \checkmark$$

For the other direction we invert. Since $\det = 19(3) - 13(4) = 5$,

$${}_{\mathcal{B}}P_{\mathcal{C}} = \left({}_{\mathcal{C}}P_{\mathcal{B}}\right)^{-1} = \frac{1}{5} \begin{bmatrix} 3 & -13 \\ -4 & 19 \end{bmatrix} = \begin{bmatrix} 3/5 & -13/5 \\ -4/5 & 19/5 \end{bmatrix}.$$

Example 4.13 (Change of Coordinates in $P_2(\mathbb{R})$)

Find ${}_{\mathcal{B}}P_{\mathcal{C}}$ where

$$\mathcal{C} = \{1,\ x,\ x^2\}, \qquad \mathcal{B} = \{1 - x^2,\ x,\ 1 + x^2\}$$

are two bases for $P_2(\mathbb{R})$. Then find a formula for $[p]_{\mathcal{B}}$ when $p = a + bx + cx^2$.

Solution. In standard coordinates, the $\mathcal{B}$ polynomials are the columns

$$1 - x^2 \to \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \qquad x \to \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad 1 + x^2 \to \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix},$$

and since $\mathcal{C}$ is the standard basis, the $\mathcal{C}$ polynomials give the identity. Row reducing the block

matrix:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ -1 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_3+R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 1 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{\frac{1}{2}R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1/2 & 0 & 1/2 \end{array}\right] \xrightarrow{R_1-R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & 0 & -1/2 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1/2 & 0 & 1/2 \end{array}\right].$$

Therefore

$${}_{\mathcal{B}}P_{\mathcal{C}} = \begin{bmatrix} 1/2 & 0 & -1/2 \\ 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \end{bmatrix}.$$

For $p = a + bx + cx^2$, the standard coordinates $[p]_{\mathcal{C}}$ are just a, b, c, so

$$[p]_{\mathcal{B}} = {}_{\mathcal{B}}P_{\mathcal{C}}\,[p]_{\mathcal{C}} = \begin{bmatrix} 1/2 & 0 & -1/2 \\ 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \frac{1}{2}(a-c) \\ b \\ \frac{1}{2}(a+c) \end{bmatrix}.$$

Check. Take $p = 3 + 2x + x^2$, so $a = 3$, $b = 2$, $c = 1$, and the formula gives the coordinates $1, 2, 2$. Indeed,

$$1(1 - x^2) + 2(x) + 2(1 + x^2) = 3 + 2x + x^2. \checkmark$$